BSc CS IT — PRACTICAL FILE

PRACTICAL / TUTORIAL NO. 10

AUTOMATION & REGRESSION ANALYSIS

Subject: Software Testing / Test Automation

Student Name	RUTVIK VALA
Enrollment Number	2504030400048
Program	BSc CS IT

Includes Excel workbook tables and the five supplied screenshot images.

Implementation

Implementation	Unnamed: 1
Parameter	Details
Project Name	Student Login & Registration System
Application/Software	Sample Web Application
Module Name	Login, Registration and Dashboard
SDLC Model	Waterfall Model (academic example)
Programming Language	HTML, CSS, JavaScript
IDE/Tool Used	Visual Studio Code, Selenium WebDriver, Chrome
Input/Test Data	Valid login, invalid password, blank fields, registration data
Student Name	Rutvik Vala
Enrollment No.	2504030400048
Aim	Analyze automation candidates and prepare a regression test suite.

Module Analysis

Module Analysis	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5
Module Name	Functionality	Critical	Frequently Changed	Suitable for Automation	Reason
Login	Authenticate user credentials	Yes	No	Yes	Stable and repetitive function
Registration	Create a new user account	Yes	Yes	Yes	Repeated validation tests can be automated
Dashboard	Display user information	Medium	No	Yes	Stable checks with clear expected output
Help/About	Display static information	No	No	No	Low-risk and rarely changed

Test Case Details

Test Case Details	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7
Test Case ID	Requirement ID	Module	Feature	Scenario	Objective	Test Type	Technique
TC-001	REQ-LOGIN-01	Login	Authentication	Valid Login	Verify valid credentials	Functional	Black-Box
TC-002	REQ-LOGIN-01	Login	Authentication	Invalid Password	Reject wrong password	Functional	Black-Box
TC-003	REQ-LOGIN-02	Login	Validation	Empty Fields	Verify required-field validation	Functional	Black-Box
TC-004	REQ-REG-01	Registration	Validation	Valid Registration	Verify account creation	Functional	Black-Box
TC-005	REQ-DASH-01	Dashboard	Page Display	Dashboard loads	Verify dashboard after login	Regression	System

Automation Candidates

Automation Candidates	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7
Test Case ID	Functionality	Criticality	Frequency	Stable	Automation Recommended	Reason	Tool
TC-001	Valid Login	High	High	Yes	Yes	Repeated, clear expected result	Selenium WebDriver
TC-002	Invalid Password	High	High	Yes	Yes	Repeated validation test	Selenium WebDriver
TC-003	Empty Fields	Medium	High	Yes	Yes	Easy to repeat and verify	Selenium WebDriver
TC-004	Valid Registration	High	Medium	Partly	Yes	Useful for regression runs	Selenium WebDriver
TC-006	Static Help Page	Low	Low	Yes	No	Low business value for automation	Manual

Test Case Execution

Test Case Execution	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6
Step No.	Test Case ID	Test Steps	Input	Expected Result	Actual Result	Status
1	TC-001	Open login page	Application URL	Login page opens	Login page opens	Pass
2	TC-001	Enter valid credentials	student / 1234	Credentials accepted	Credentials accepted	Pass
3	TC-001	Click Login	Login button	Dashboard opens	Dashboard opens	Pass
1	TC-002	Enter wrong password	student / wrong	Error message	Error message	Pass
1	TC-003	Submit empty fields	blank / blank	Required-field message	Required-field message	Pass
1	TC-004	Enter valid registration details	Valid user details	Account created	Account created	Pass

Regression Test Suite

Regression Test Suite	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4
Regression Test ID	Module	Scenario	Expected Result	Status
RS-001	Login	Valid credentials	Login Successful	Pass
RS-002	Login	Invalid password	Invalid Credentials	Pass
RS-003	Login	Empty fields	Fields are required	Pass
RS-004	Registration	Valid registration	Account created	Pass
RS-005	Dashboard	Load dashboard after login	Dashboard displayed	Pass

Automation Feasibility

Automation Feasibility	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4
Test Case ID	Functionality	Recommended	Reason	Tool
TC-001	Valid Login	Yes	Stable and frequently executed	Selenium WebDriver
TC-002	Invalid Password	Yes	Repeated validation scenario	Selenium WebDriver
TC-003	Empty Fields	Yes	Simple repeatable validation	Selenium WebDriver
TC-004	Valid Registration	Yes	Useful for repeated regression runs	Selenium WebDriver
TC-006	Static Help Page	No	Low priority and low repetition	Manual

Regression Summary

Regression Summary	Unnamed: 1
Metric	Value
Total Modules Analyzed	4
Automation Candidates	4
Regression Test Cases	
Passed Test Cases	
Failed Test Cases	
Pass Percentage	
Recommendation	Automate stable, repetitive and high-priority tests
Overall Result	

Output Screenshots

Tutorial 10 — Illustrative Output Screenshots
Note: Illustrative mockups, not screenshots captured from a live application.
Output 1: 01_module_analysis.png
Output 2: 02_automation_candidate.png
Output 3: 03_regression_suite.png
Output 4: 04_feasibility_report.png

MCQ Answers

MCQ Answers	Unnamed: 1	Unnamed: 2
Q. No.	Correct Answer	Explanation
1	C) Black-Box Testing	Verifies functionality without examining source code.
2	D) White-Box Testing	Requires knowledge of internal program code.
3	B) Communication between modules	Integration testing checks module interaction.
4	C) Complete Application Behavior	System testing validates the complete application.
5	B) White-Box Testing	Checks execution paths and branch coverage.

Conclusion & Viva

Conclusion & Viva	Unnamed: 1
Item	Answer
Conclusion	Automation and regression analysis identify repeatable test cases and help verify existing features after software changes.
What is test automation?	Using tools or scripts to execute test cases automatically.
What is regression testing?	Retesting existing features after software changes.
Name an automation tool.	Selenium WebDriver.
Good automation candidates?	Stable, repetitive, high-priority tests with clear expected results.

SCREENSHOT EVIDENCE

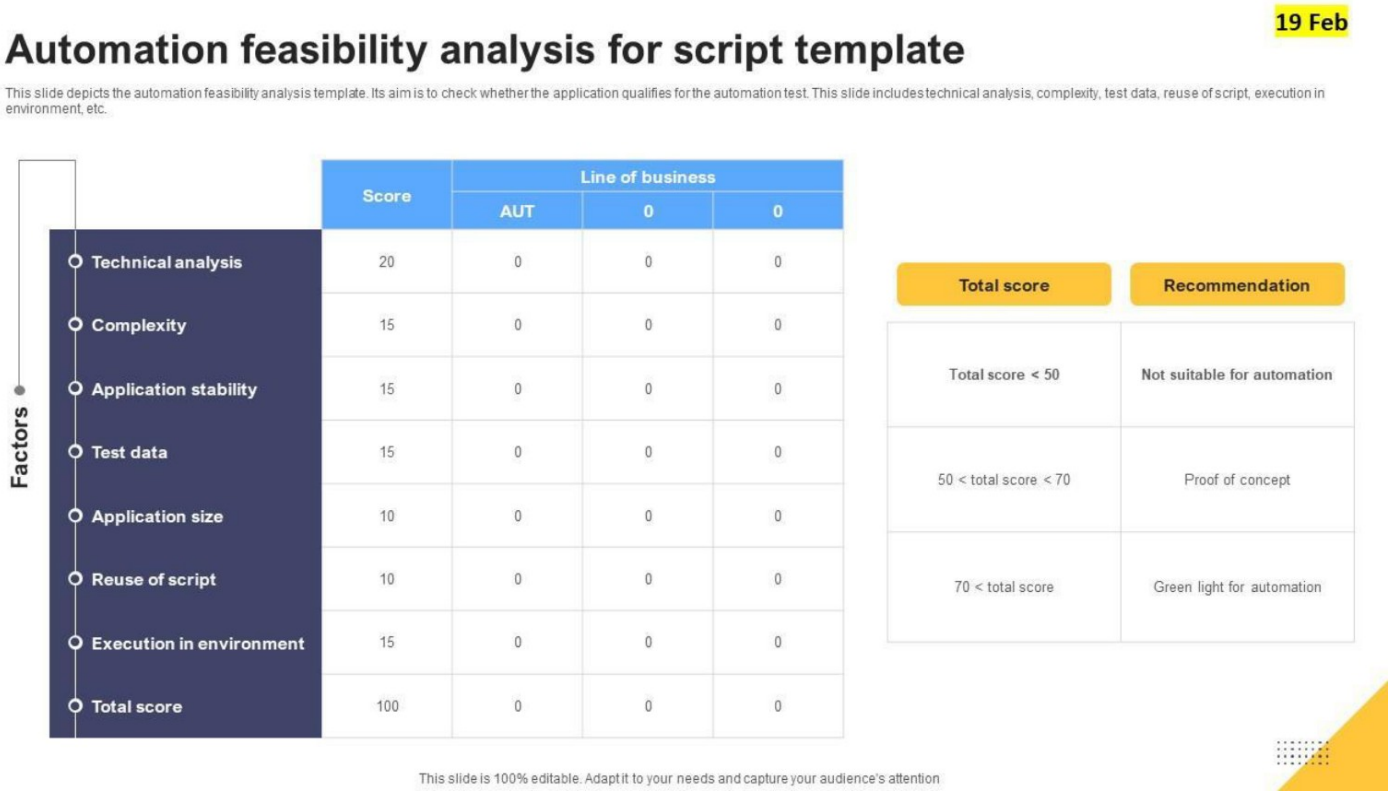
Screenshot 1 — Automation Test Case Selection (Squash)



Screenshot supplied for practical reference/evidence.

SCREENSHOT EVIDENCE

Screenshot 2 — Automated Login Test Execution Log



19 Feb

Automation feasibility analysis for script template

This slide depicts the automation feasibility analysis template. Its aim is to check whether the application qualifies for the automation test. This slide includes technical analysis, complexity, test data, reuse of script, execution in environment, etc.

Factors

	Score	Line of business		
		AUT	0	0
Technical analysis	20	0	0	0
Complexity	15	0	0	0
Application stability	15	0	0	0
Test data	15	0	0	0
Application size	10	0	0	0
Reuse of script	10	0	0	0
Execution in environment	15	0	0	0
Total score	100	0	0	0

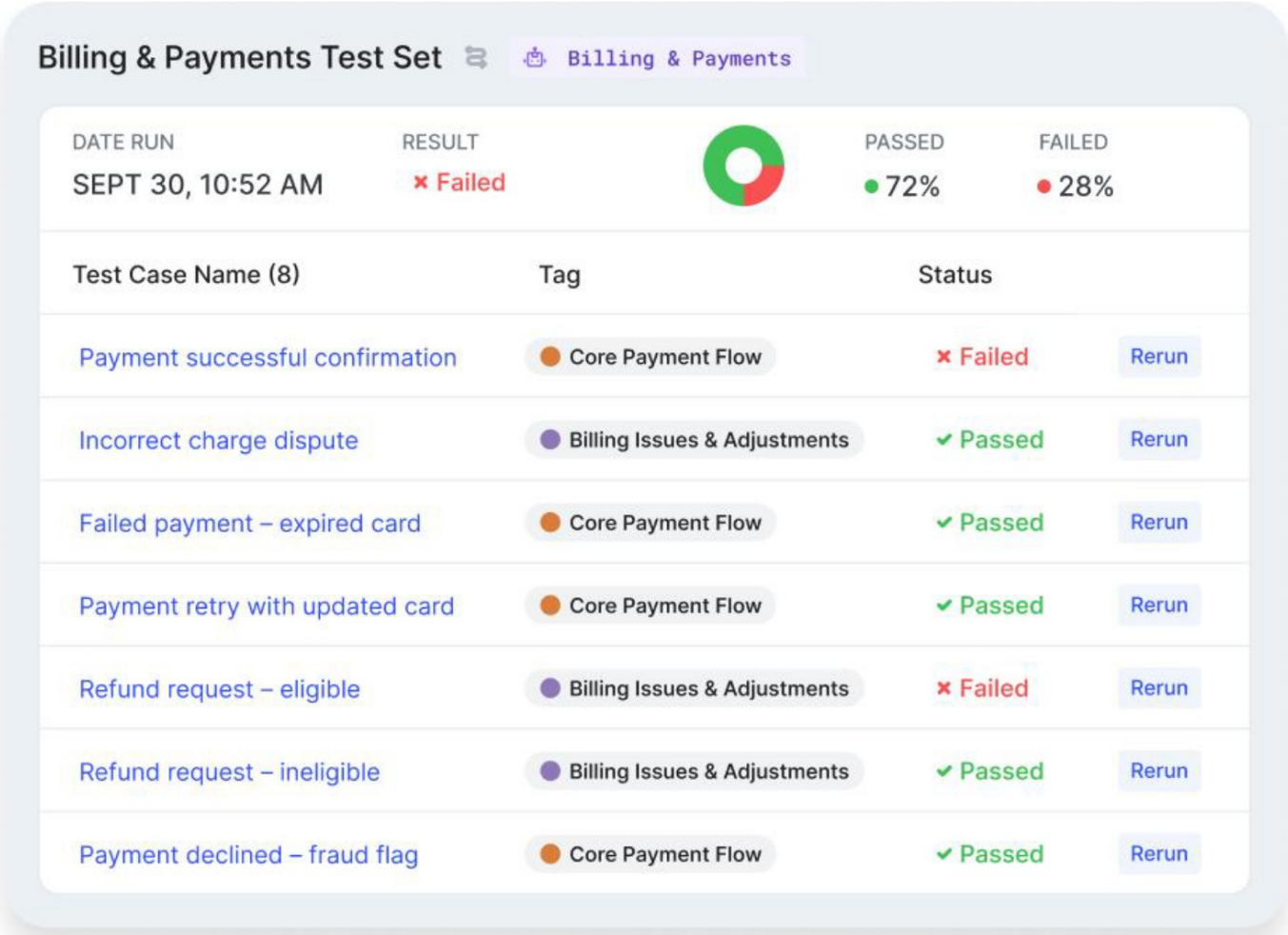
Total score	Recommendation
Total score < 50	Not suitable for automation
50 < total score < 70	Proof of concept
70 < total score	Green light for automation

This slide is 100% editable. Adapt it to your needs and capture your audience's attention

Screenshot supplied for practical reference/evidence.

SCREENSHOT EVIDENCE

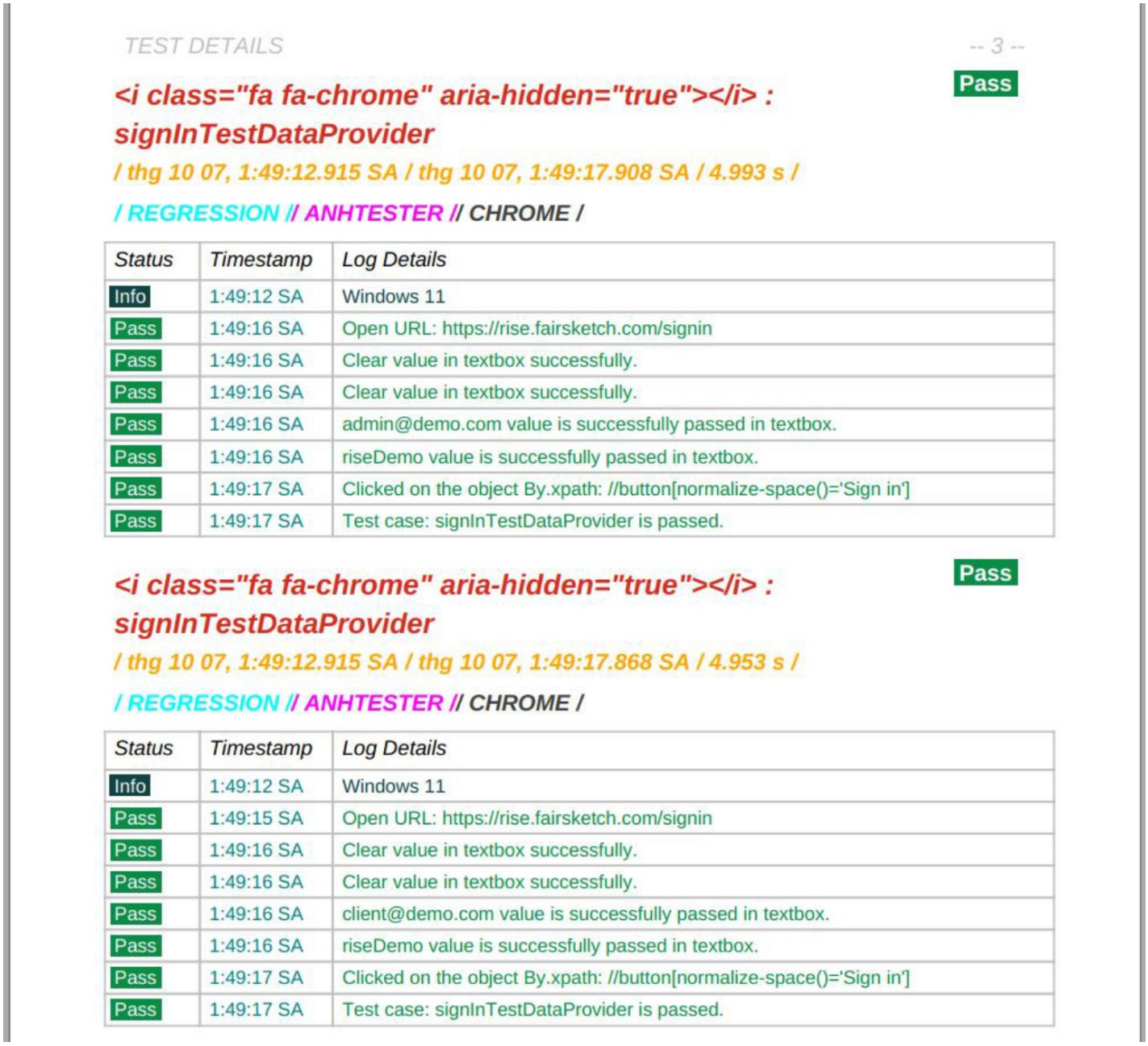
Screenshot 3 — Billing & Payments Test Set Results



Screenshot supplied for practical reference/evidence.

SCREENSHOT EVIDENCE

Screenshot 4 — Automation Feasibility Analysis



TEST DETAILS -- 3 --

** : signInTestDataProvider** Pass

/ thg 10 07, 1:49:12.915 SA / thg 10 07, 1:49:17.908 SA / 4.993 s /

/ REGRESSION // ANHTESTER // CHROME /

Status	Timestamp	Log Details
Info	1:49:12 SA	Windows 11
Pass	1:49:16 SA	Open URL: https://rise.fairsketch.com/signin
Pass	1:49:16 SA	Clear value in textbox successfully.
Pass	1:49:16 SA	Clear value in textbox successfully.
Pass	1:49:16 SA	admin@demo.com value is successfully passed in textbox.
Pass	1:49:16 SA	riseDemo value is successfully passed in textbox.
Pass	1:49:17 SA	Clicked on the object By.xpath: //button[normalize-space()='Sign in']
Pass	1:49:17 SA	Test case: signInTestDataProvider is passed.

** : signInTestDataProvider** Pass

/ thg 10 07, 1:49:12.915 SA / thg 10 07, 1:49:17.868 SA / 4.953 s /

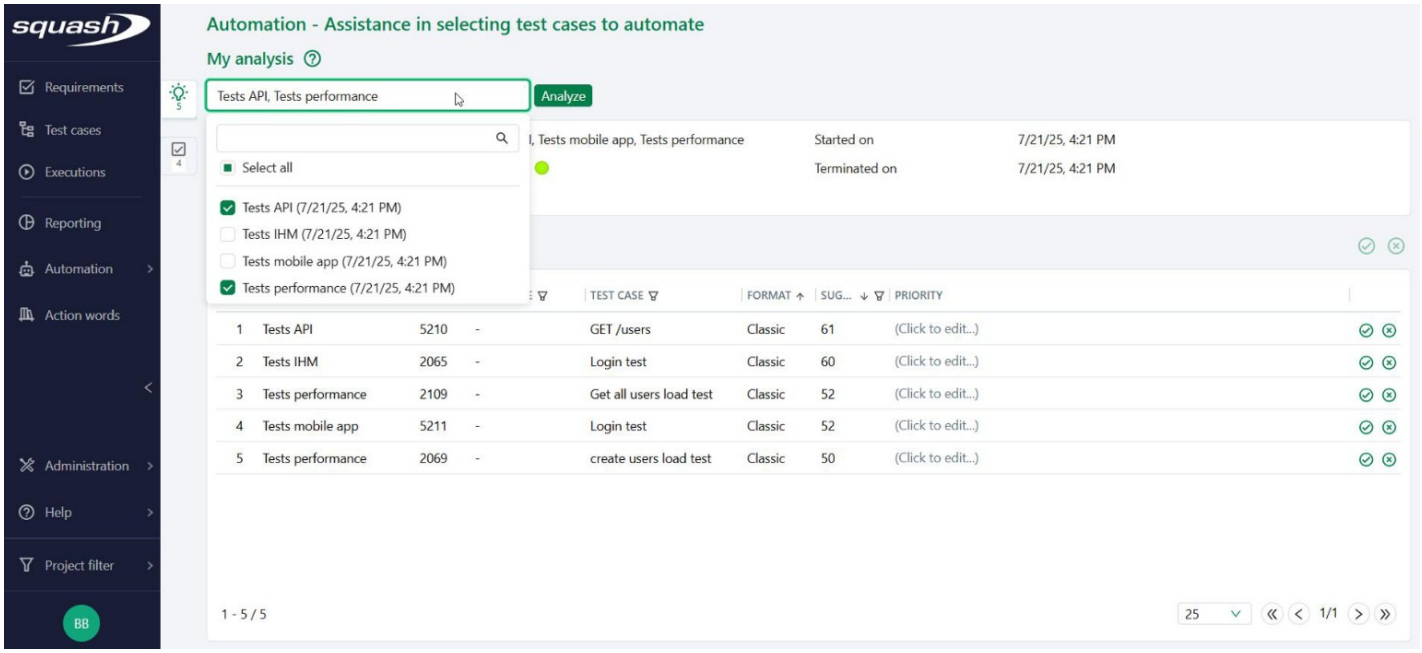
/ REGRESSION // ANHTESTER // CHROME /

Status	Timestamp	Log Details
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Pass	1:49:15 SA	Open URL: https://rise.fairsketch.com/signin
Pass	1:49:16 SA	Clear value in textbox successfully.
Pass	1:49:16 SA	Clear value in textbox successfully.
Pass	1:49:16 SA	client@demo.com value is successfully passed in textbox.
Pass	1:49:16 SA	riseDemo value is successfully passed in textbox.
Pass	1:49:17 SA	Clicked on the object By.xpath: //button[normalize-space()='Sign in']
Pass	1:49:17 SA	Test case: signInTestDataProvider is passed.

Screenshot supplied for practical reference/evidence.

SCREENSHOT EVIDENCE

Screenshot 5 — Module Analysis for Sample Web App



Screenshot supplied for practical reference/evidence.